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PROJECTS

Gemma 4 Inference Megakernel

GitHub repo

- Writing gemma.c, a Gemma 4 megakernel written in CUDA and PTX. Full forward pass is one kernel.

Sinter

GitHub repo

- Sinter is a search-based megakernel compiler and inference engine written in C++ and Rust, aiming for something between Luminal and Slang.
- We define 13 semantic ops, such as GEMM, softmax, and convolution, allowing us to compose complex architectural primitives. Each semantic op has chip-specific lowerings, in which we also specify how lowerings interact via smem tiles, gmem counters, and mbarriers.
- Sinter emits a standalone inference engine repository for the selected hardware configuration, making it much easier to hillclimb performance and creating much more hackable and digestible code.
- In Rust, we implement the core serving shell, including tokenization, continuous batching, and paged and radix attention, while C++ owns CUDA and the compiler.

Writing Kernels

- Writing custom GPU kernels in CUDA, PTX, and ThunderKittens, including flash attention, KDA, LatentMOE, etc.

Snaptokens

Blog post

GitHub repo

- Built a tokenizer in rust. Achieves exact token-ID and API-level parity, while being 46.41x faster than Hugging Face and 2.19x faster than Gigatoken on inference-shaped workloads. Current fastest OSS tokenizer in the world.

LPT

GitHub repo

- Built a latent planning token path that injects planning tokens every n rollouts, then adds their final-layer hidden states to the next n rollouts.
- Improved perplexity by 4.6% for 0.3% overhead in a matched run; the mechanism should become more useful with scale.

STRATUS UAV: Optimizing Prescribed Burning using UAVs and Predictive Modeling

- Led development of a patent-pending autonomous UAV to mitigate wildfire spread; 3D-printed a VTOL aircraft for autonomous backburning with 15 km range and 1.2 hour endurance.
- Targeted core field constraints: helicopter operations around \$16k/day, custom aircraft and pilots, and pilot liability; designed for 5x lower price and 3x higher range than the closest drone competitor.

SBON (Strategic-Blackline-Optimization-Engine)

GitHub repo

- Built a wildfire prediction model that set state-of-the-art results and created the first ML approach to optimizing prescribed burns.
- Built a tri-branch multimodal architecture across 13 input modalities spanning fire imagery, weather, and geography; ConvNeXt and SSM encoders feed a larger SSM for 4-hour fire-mask prediction.
- Treated the model as an environment, rolled out predictions to 48 hours, and used PPO to optimize burn placement in a niche field with fewer than 20 papers in the last 30 years.

The no-circles project

no-circles.com | GitHub repo

- Built a personalized daily newsletter that learns from each reader's replies and behavior; scaled to triple-digit users within 3 weeks.
- Built the user memory, daily brief, and reply feedback loop to explain unfamiliar fields through topics each reader already cares about.

PHOENIX UAV / Science Fair Results

GitHub repo

- Built PHOENIX UAV for early wildfire detection; placed 2nd in the California State Science Fair, 1st at regionals, and won 5 awards including the Henkel aerospace award and Junior Lemelson-MIT Inventor Award.

EDUCATION

Astra Nova High School

2026-2028

Independent Study / Competitions

- Selected for SPARC '26 (3% acceptance rate); achieved USACO Gold and F=MA qualifying scores in the top 5% nationally.
- Competed in CALICO, Harker Physics Invitational, and Harker Programming Invitational; held a top 30 national placement in MSPF Debate with 2 tournament wins and elimination rounds at Stanford and TOC.